AIDCF-007A Control Universe Constitutional Amendment

15 / 90 / 360 Baseline, with v1.1 amendment note AIDCF-007A-AMD-002

Document control

Field	Value
Public ID	AIDCF-007A (amendment instrument)
Amendment records	AIDCF-007A-AMD-001 (v1.0, cardinality and identifiers); AIDCF-007A-AMD-002 (v1.1, control content, gate mapping and assurance levels)
Version	1.1
Date	2026-10-10
Status	Draft for controlled review. AMD-001 remains in force as the cardinality and identifier baseline (see §10); AMD-002 is proposed (see §11).
Author	Siva Sethumadhavan
Amends and supersedes	AIDCF-007 Master Control Framework and Control Catalogue v1.0 (archived at `archive/superseded/AIDCF-007_Control_Catalogue_v1.0.md` and `.xlsx`)
Governs	AIDCF-007A Canonical Master Control Registry (v1.0 in force; v1.1 Proposed)
Downstream effect	Binding on AIDCF-008 to AIDCF-019, AIDCF-VR-001 and AIDCF-WP-001 for control identity, cardinality and gate mapping
Control universe	15 domains; 90 control families; 360 normative controls
Deprecated statement	14 domains; 84 families; 336 controls (narrative statements only; see §2)
Registry	`standards/aidcf-007a/registry/AIDCF-007A_Canonical_Master_Control_Registry_15-90-360_v1.0.xlsx` (v1.0); registry v1.1 per §11

Revision history

Version	Date	Summary
1.0	Not stated	AIDCF-007A-AMD-001 issued: fixed 15/90/360 cardinality and MCF identifiers; deprecated 14/84/336 statements.
1.1	2026-10-10	Clarified that the amendment supersedes the AIDCF-007 v1.0 catalogue and governs the AIDCF-007A registry; corrected the change history; added approval record, CG-01 to CG-15 names table and amendment note AIDCF-007A-AMD-002 (registry v1.1 proposals).

1. Constitutional decision

The authoritative AIDCF control universe consists of 15 control domains, 90 control families and 360 normative controls. Each domain contains six control families and 24 controls. Each family contains four controls. This decision is the sole cardinality baseline for the AIDCF-007A registry and all dependent artifacts.

This amendment amends and supersedes the AIDCF-007 Master Control Framework and Control Catalogue v1.0. AIDCF-007 v1.0 is archived and retained for lineage only. All live references to the control universe shall use the

public ID AIDCF-007A; "AIDCF-007" (without the suffix) denotes only the superseded v1.0 catalogue.

2. Supersession and corrected change history

The record of how the cardinality conflict arose is as follows.

1. The AIDCF-007 v1.0 workbook (`AIDCF-007_Control_Catalogue_v1.0.xlsx`, Summary sheet) already declared and contained 15 domains, 90 control families and 360 normative controls, using identifiers MCF-<DOM>-001 to MCF-<DOM>-024.
2. Separately, some narrative AIDCF documents stated a 14-domain / 84-family / 336-control universe. The conflict between these narrative statements and the workbook was recorded as a blocker in the full-stack validation (AIDCF-VR-001) and in the AIDCF-015 pilot, which used local pilot identifiers instead of MCF identifiers because of it.
3. AIDCF-007A-AMD-001 resolved the conflict in favour of the workbook. It did not add a domain or renumber any control; it confirmed the 15/90/360 universe that the workbook already held and deprecated the narrative 14/84/336 statements.

Every statement declaring 14 domains, 84 families or 336 controls is deprecated and superseded. Such statements shall not be used for coverage, scoring, conformance, reporting, machine execution or assurance decisions.

3. Canonical identifiers

The identifiers MCF-<DOMAIN>-001 through MCF-<DOMAIN>-024 are stable. Family identifiers use <DOMAIN>-01 through <DOMAIN>-06; family *n* holds controls 4*n*−3 to 4*n*. Identity is immutable; a meaning-changing replacement requires a new identifier and an explicit supersession record. Title corrections that remove ambiguity without changing meaning (see §11.2) are recorded as amendments and do not change the identifier.

4. Domain model

Code	Domain	Families	Controls	v1.0 primary gates (superseded by §11.3)
GOV	Governance, Accountability and Assurance Direction	6	24	CG-13; CG-14
CAP	AI Capability, Service Outcome and Demand Management	6	24	CG-01; CG-02
DEP	Dependency, Concentration and External Reliance Management	6	24	CG-03; CG-12
PHY	Physical Site, Building and Security Infrastructure	6	24	CG-02; CG-04
POW	Electrical Power Infrastructure and Energy Continuity	6	24	CG-05
THM	Cooling, Thermal and Fluid Infrastructure	6	24	CG-06
ENV	Environmental and Resource Performance	6	24	CG-02; CG-04
CMP	AI Compute Infrastructure	6	24	CG-08
STO	AI Storage and Data Infrastructure	6	24	CG-08; CG-10
NET	AI Network and Interconnect Architecture	6	24	CG-08

Code	Domain	Families	Controls	v1.0 primary gates (superseded by §11.3)
OPS	Building Operations and Service Management	6	24	CG-07; CG-13; CG-14
RES	Resilience, Continuity and Recovery	6	24	CG-09; CG-10
SUP	Supply Chain, OEM and Ecosystem Assurance	6	24	CG-03; CG-13
SEC	Digital, OT and Control-Plane Security	6	24	CG-07; CG-15
EVA	Evidence, Assessment and Assurance Management	6	24	CG-11; CG-12

The right-hand column records the whole-domain gate association used in AIDCF-007 v1.0 and registry v1.0. It is retained for lineage. Under AMD-002 it is replaced by control-specific mapping (§11.3) and shall not be used to evaluate a gate.

5. Critical gates

The critical-gate names below are those defined in AIDCF-007 v1.0 (Critical Gates sheet) and are canonical. Gate criteria, roll-up and decision ceilings are defined in AIDCF-010; evidence floors in AIDCF-008.

Gate	Name	What it protects
CG-01	Capability and outcome definition	Capability Instance, Service Outcomes, tolerances, criticality and decision question are defined, owned and approved
CG-02	Boundary and architecture integrity	Assessment boundary, current architecture, as-built reconciliation and exclusions
CG-03	Critical dependency visibility	Material internal and external dependencies, common modes and concentration are identified and owned
CG-04	Life safety and hazardous energy	Fire, arc flash and shock, hazardous fluids and chemicals, confined space, refrigerants, battery thermal runaway, safe work
CG-05	Resilient electrical supply	Utility, distribution, UPS/BESS, generation, fuel, protection and power quality across operating states
CG-06	Thermal continuity	Heat-removal path (air, liquid, facility water, heat rejection) across operating states, including transients
CG-07	Critical control-system integrity	BMS, EPMS, CDU, PLC, SCADA and automation function safely, fail safe, and are configured and governed
CG-08	Compute, storage and network functional integrity	Accelerators, storage and fabrics deliver required function, performance and integrity
CG-09	Failure containment	Fault domains, isolation and blast-radius limits prevent propagation beyond tolerance
CG-10	Recovery readiness	Restoration of outcome (power, cooling, compute, data and checkpoints, network, control plane) is planned and tested
CG-11	Evidence sufficiency	Evidence for decisive claims meets floors (meta-gate, evaluated across all controls)
CG-12	Critical UNKNOWN	No unresolved US4/US5 UNKNOWN without disposition (meta-gate, evaluated across all controls)
CG-13	Responsibility and authority	Every critical function and dependency has an accountable owner and decision authority
CG-14	Change control	Material changes are controlled, impact-assessed and validated

Gate	Name	What it protects
CG-15	Integrated digital authority over infrastructure	Identity, privilege, remote access and control-plane reach over facility and AI infrastructure are bounded, monitored, recoverable

6. Registry authority

The registry workbook is the canonical machine-readable form of the control universe. It contains the domain, family and control inventories and the APS, gate, ontology, failure, graph, evidence, assessment, assurance-level, lifecycle, UNKNOWN and reassessment mappings. Where narrative text and the registry conflict, this amendment and the controlled registry govern cardinality and identifiers. Substantive control changes require controlled methodology review and are recorded as numbered amendments (AIDCF-007A-AMD-NNN).

7. Normative governance rules

- No domain, family or control may be deleted from an approved baseline.
- Records may be ACTIVE, DEPRECATED or SUPERSEDED.
- Profiles may strengthen or add requirements but may not silently weaken the baseline.
- NOT_APPLICABLE requires approved rationale; missing evidence is UNKNOWN.
- Critical-gate status remains separate from scores and maturity.
- All control changes require impact analysis across AIDCF-008 to AIDCF-019 and the AIDCF-005 AITG mappings.
- Testing remains authorised, competent, proportionate and safe.

8. Downstream migration

Artifact	Required action
AIDCF-008	Reconcile control-to-evidence profiles to 360 controls and the per-control evidence profiles of registry v1.1.
AIDCF-009	Confirm 360 procedures and 90 family workbooks use canonical IDs and the per-control procedures of registry v1.1.
AIDCF-010	Apply per-control CRITICAL/SUPPORTING gate mapping and the canonical gate roll-up rule.
AIDCF-011	Update programme profiles and coverage reporting; align required assurance levels to AL2–AL5.
AIDCF-012	Update report disclosures and control coverage registers.
AIDCF-013	Reconcile domain architecture control links to AIDCF-007A family ranges.
AIDCF-014	Publish corrected registries, vocabularies and migration record (serialisation per AIDCF-016).
AIDCF-015	Replace pilot identifiers with canonical MCF identifiers and close the related baseline issue after validation.

9. Acceptance tests

- Exactly 15 unique domain codes.
- Exactly 90 unique family IDs, six per domain.
- Exactly 360 unique control IDs, 24 per domain and four per family.
- No duplicate or orphan control or family identifiers.
- No two controls share the same title (added by AMD-002).

- Every control has APS, critical-gate, ontology, failure, graph, evidence, assessment, UNKNOWN and reassessment mappings.
- Every control has status and version.
- Under registry v1.1: every gate CG-01 to CG-10 and CG-13 to CG-15 has at least three CRITICAL controls; CG-11 and CG-12 have no per-control mapping; every control's required assurance level is consistent with its claim type (§11.4).

10. Approval record

Item	Record
Instrument	AIDCF-007A-AMD-001 (cardinality 15/90/360 and canonical identifiers)
Approved by	Siva Sethumadhavan, author, acting as baseline owner
Approval date	As stated on the v1.0 issue. The v1.0 document records the status "Approved baseline" but states no separate approval date.
Scope of approval	Cardinality and identifiers only. It is not approval of control wording, mappings, evidence profiles or any engineering content.
Independent multidisciplinary ratification	PENDING. No independent reviewer, committee or external body has ratified AMD-001.
Effect while ratification is pending	AMD-001 is the working baseline for identity and cardinality across AIDCF artifacts. It creates no certification, engineering approval or assurance by itself.
Instrument	AIDCF-007A-AMD-002 (§11)
Status	Draft for controlled review. Not approved.

11. Amendment note AIDCF-007A-AMD-002 (v1.1)

Field	Value
Amendment ID	AIDCF-007A-AMD-002
Date	2026-10-10
Status	Draft for controlled review
Subject	AIDCF-007A Canonical Master Control Registry v1.1
Registry status	Proposed — subject to controlled methodology review
Identity effect	None. All 15 domain codes, 90 family IDs and 360 control IDs are unchanged.

11.1 Control content rewritten per control

Registry v1.0 (and AIDCF-007 v1.0) generated every control's objective, requirement, assessment criteria, evidence and procedure text from a single template ("Ensure … is defined, implemented, maintained and demonstrably supports …"), with only the title substituted. Registry v1.1 replaces that template text with content specific to each control:

- objective and one or more "shall" requirements naming the accountable role;
- three to seven independently testable assessment criteria;

- parameters, expressed as capability-profile parameters with illustrative defaults where values are site-specific;
- an evidence profile per control (claim type, profile tier CP-1 to CP-5, required evidence classes E1–E7, quality and sufficiency floors, operating-state coverage, independence, UNKNOWN effect and reassessment triggers) aligned to AIDCF-008;
- an ordered assessment procedure using the canonical assessment methods, aligned to AIDCF-009;
- control type, lifecycle stages, operating states, applicability, failure modes, ontology classes, testing-safety constraints, a control-specific UNKNOWN rule, informative references and related controls.

Control IDs, family IDs, family membership and family structure are unchanged.

11.2 Duplicate titles corrected

Two control titles duplicated titles elsewhere in the registry. They are disambiguated without change of meaning or identifier:

Control ID	v1.0 title	v1.1 title	Duplicated with
MCF-NET-004	Fault-Domain Design	Fabric Fault-Domain Design	MCF-CMP-008 Fault-Domain Design
MCF-STO-019	Recovery Sequencing	Storage Recovery Sequencing	MCF-RES-014 Recovery Sequencing

11.3 Control-specific gate mapping

Registry v1.0 mapped every control in a domain to the domain's primary gates (§4), so that, for example, all 24 STO controls were mapped to CG-10. That mapping did not distinguish controls whose failure defeats a gate from controls that merely inform it. Registry v1.1 maps each control individually, 0..n gates per control, each with a role:

- **CRITICAL**: the gate cannot PASS unless this control is CONFORMING at its required claim type, when in scope.
- **SUPPORTING**: the control informs the gate evaluation; a NONCONFORMING result requires a documented gate-level evaluation but does not automatically fail the gate.

Mappings are made by meaning and across domains (for example, CG-04 draws on relevant PHY, POW, THM and ENV controls; CG-10 on POW, THM, CMP, STO, NET and RES recovery controls). CG-11 and CG-12 are meta-gates evaluated across all controls and are not mapped per control. Each of CG-01 to CG-10 and CG-13 to CG-15 has at least three CRITICAL controls. The gate roll-up rule is defined in AIDCF-010.

11.4 Assurance levels assigned per control

Registry v1.0 assigned required assurance levels A2 to A5 in rotation by position within each family. Registry v1.1 assigns each control a primary claim type and the minimum assurance level that claim type requires:

Claim type	Minimum assurance level
DESIGN	AL2
IMPLEMENTATION	AL3
OPERATION	AL4
EFFECTIVENESS	AL5
RESILIENCE	AL5

Legacy codes A2 to A5 map one-to-one to AL2 to AL5.

11.5 Gate names restored

Some downstream documents had shortened or reworded critical-gate names (for example "Electrical continuity" for CG-05). Registry v1.1 and all dependent artifacts use the gate names from AIDCF-007 v1.0, as listed in §5.

11.6 Status and conditions

Registry v1.1 is Proposed — subject to controlled methodology review. Until AMD-002 is approved, registry v1.0 remains the identity and cardinality baseline; v1.1 content may be used for pilots, review and calibration, and any assessment that uses it shall disclose the registry version. Approval of AMD-002 requires the multidisciplinary review set out in §12.

12. Controlled-review caveat

The cardinality and identifiers are fixed by AMD-001. The adequacy of individual control wording, evidence profiles, gate mappings and assurance levels under AMD-002 remains subject to multidisciplinary controlled review (governance and assurance methodology, electrical, mechanical and liquid cooling, AI compute, storage and network architecture, operations, OT and security, evidence), calibration and pilot evidence. This amendment does not create certification, engineering approval or assurance by itself.